WAWIRA JOSHUA NYAGA

TDC113-C002-0003/2024

COMPUTER SCIENCE – LEVEL 6

BASIC ELECTRONICS

CT/CU/CS/CC/01/6/8

1 ½ HOURS

JULY 2025

JKUAT TVET INSTITTUTE – KAREN CAMPUS
WRITTEN ASSESSMENT
CAT 1

Answer ALL questions

SECTION A (20 Marks)

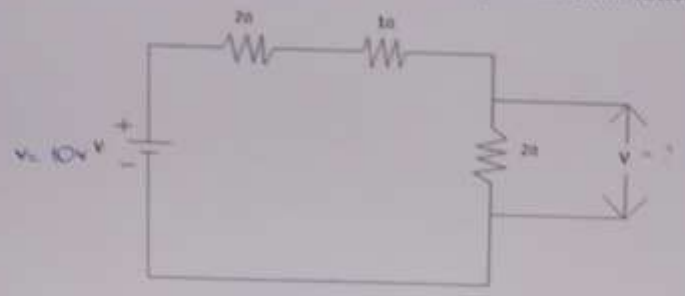
Each question contains (4 marks)

1. Electrical circuits are of two different types: direct current circuits and alternating current circuits. Highlight TWO differences between the two types of circuits. (4 marks)
2. Draw the structure of an atom and label all the components. (4 marks)
3. Draw a simple circuit consisting of a cell, a switch and two registers in parallel. (4 marks)
4. State the following laws as used in electronics
   a). Thevenin theorem. (2 marks)
   b). Kirchhoff's Current Law. (2 marks)
5. State four standard electrical quantities and the corresponding formula for calculating them. (4 marks)

SECTION B (30 Marks)

6. a). Describe four number systems used in computing. (8 marks)
   b). Convert 79_{10} binary (4 marks)
   c). Convert $1BF_{16}$ to octal. (4 marks)
   d). Convert 107_8 to hexadecimal. (4 marks)

7. a). Study the circuit diagram below and answer the question that follow.



Calculate Voltage across 2Ω Resistor where supply v= 10volts. (6 marks)

b) Current of 1.4A when flowing through a circuit for 15 minutes dissipates 200 KJ of energy. Calculate: (4 marks)

i) the p.d;

ii) power dissipated;

iii) the resistance of the circuit.

44/50

JKUAT CONTINOUS ASSESSEMENT TESTS

NAME: WAWIRA JOSHUA NYAGA TVET

REG NO: T0113-1002-0025/2024

COURSE: Computer Science YEAR OF STUDY: 1

SHEET NO:

UNIT CODE: TITLE: Basic Electronics

DATE: 10/7/2025

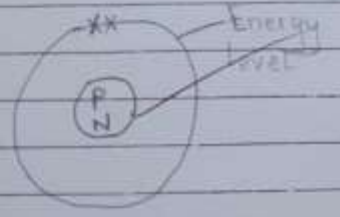
NOTE: This stationery will be used for Continuous Assessment work only

It will be a breach of examination regulations to use it otherwise.

SECTION A

1. Direct currents circuits are circuits which contain the amount of power they can produce by a generator while alternating current circuits are circuits which they use to supply the required power supply to the system

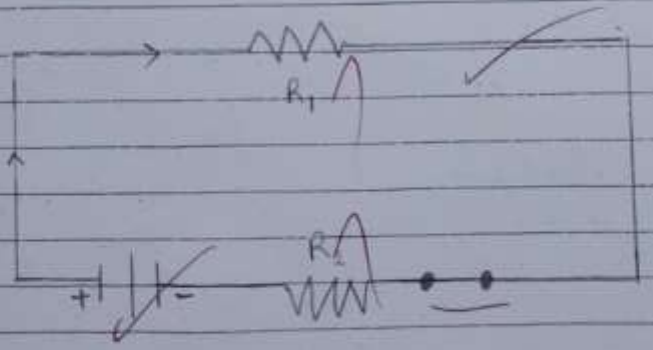
2.



Key
xx - Electrons
P - Protons
N - Neutrons

08

3.



02

06

4 a) Thevenin theorem states that complex circuit can be reduced into one simple equivalent circuit with one single volt and a resistance in series

b) Kirchhoff's law states that the total current entering a junction must be equal to the total current leaving the junction.

08

5. i) Voltage - measured in Volts (V)

$V = I \times R$

ii) Current - measured in Amperes (A)

$I = \frac{V}{R}$

iii) Resistance - measured in Ohms (Ω)

$R = \frac{V}{I}$

08

iv) Power - measured in Watts (W)

$P = V \times I$ $P = V \times I$

SECTION B

6a) i) Binary Number System - is a number system with a base 2, with symbols 0 and 1

ii) Octal Number System - is a number system with a base 8, with symbols 0, 1, 2, 3, 4, 5, 6, 7

iii) Hexadecimal Number System - is a number system with a base 16 with symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.

iv) Decimal Number System - is a number system with a base 10, with symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

08

b) Convert 79_{10} to binary

2	79
2	39 R1
2	19 R1
2	9 R1
2	4 R1
2	2 R0
2	1 R0
	0 R1

$79_{10} = 1001111_2$

08

20

c) Convert $1BF_{16}$ to Octal

A-1010

B-1011

C-1100

D-1101

E-1110

F-1111

1 = 0001

B = 1011

F = 1111

110 = $(2^2 \times 1) + (2^1 \times 1) + (2^0 \times 0) = 6$

111 = $(2^2 \times 1) + (2^1 \times 1) + (2^0 \times 1) = 7$

0001 1011 1111

0 6 7 7

$= 677_8$

04

d) Convert 107_8 to hexadecimal

$1 = 001_2$

$0 = 000_2$

$7 = 111_2$

~~0100~~

100 = $(2^2 \times 1) + (2^1 \times 0) + (2^0 \times 0) = 4$

111 = $(2^2 \times 1) + (2^1 \times 1) + (2^0 \times 1) = 7$

0000 1000 0111

0 4 7

$= 47_{16}$

04

7a) $R_T = R_1 + R_2$

$R_T = 2 + 1 = 3\Omega$

$R_T = R_1 + R_2$

$R_T = 3 + 2$

7a) $R_T = R_1 + R_2 + R_3$

$2 + 1 + 2$

$R_T = 5\Omega$

$I = \frac{V}{R}$

$I = \frac{10}{5} = 2A$

$V = I \times R$

$V = 2 \times 2 = 4V$

$= 4V$

06

14